Electromagnetic Potentials

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

IRA A. FULTON SCHOOLS OF engineering
ARIZONA STATE UNIVERSITY

Lecture Objective

- To introduce the electric scalar and magnetic vector potentials for electrostatic and magnetostatic as well as electrodynamic cases.

Electromagnetic Potential Functions

- Generally speaking, a potential function is a function whose derivative yields a field.
- In physics, fields are associated with forces (e.g., Coulomb's law of force for electric field, Ampere's law of force for magnetic fields), and potentials are associated with energy.
- Often, it is 'easier' to derive the potential function and find the field rather than derive the field directly from a source distribution.

Electrostatic Potential (1 of 2)

- Since the electrostatic field is *conservative*, it can be written as the gradient of a scalar field (the electrostatic potential):

$$\nabla \times \vec{E} = 0 \quad \Rightarrow \quad \vec{E} = -\nabla V$$

Electrostatic Potential (2 of 2)

- The ***reference point*** (P_0) is where the potential is zero (***<u>ground</u>*** in circuit theory).
- Often the reference is taken to be at infinity so that the potential of a point in space is defined as

$$V(P) = -\int_{\infty}^{P} \vec{E} \cdot d\vec{l}$$

- Electric potential has units of volts.

Question

- Why do we have the freedom (or perhaps the need) to define an arbitrary reference point where the potential is zero?
- Hint: the reason is mathematical.

Poisson's and Laplace's Equations

- ***Poisson's equation*** is a differential equation for the electrostatic potential V. Poisson's equation and the boundary conditions applicable to the particular geometry form a boundary-value problem that can be solved either analytically for some geometries or numerically for any geometry.
- In many situations, instead of evaluating the electric field directly, it is easier to solve Poisson's equation for the electrostatic potential and then the electric field.

Derivation of Poisson's Equation in Free Space

$$\nabla \cdot \vec{E} = \frac{\rho_V}{\varepsilon_0}$$

$$\vec{E} = -\nabla V \quad \rightarrow \quad \nabla \cdot \nabla V$$

$$= -\frac{\rho_V}{\varepsilon_0}$$

$$\nabla^2 V$$

$$\boxed{\nabla^2 V = -\frac{\rho_V}{\varepsilon_0}}$$

Laplace's Equation

- ***Laplace's equation*** is the homogeneous form of ***Poisson's equation***.
- We use Laplace's equation to solve problems where potentials are specified on conducting bodies (boundary conditions), but no charge exists in the free space region.

$$\nabla^2 V = 0$$

Laplace's equation

Magnetostatic Potential

- Since the magnetic field is solenoidal, it can be written as the curl of another vector field (the vector magnetic potential):

$$\nabla \cdot \vec{B} = 0 \quad \Rightarrow \quad \vec{B} = \nabla \times \vec{A}$$

- Vector magnetic potential has units of Wb/m.

Dynamic (Time-Varying) Case (1 of 2)

- In the time-varying case, we have Faraday's law:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

- The electric field is no longer conservative. However, the magnetic flux density is still solenoidal.

Dynamic (Time-Varying) Case (2 of 2)

- Hence,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\frac{\partial \left(\nabla \times \vec{A}\right)}{\partial t}$$

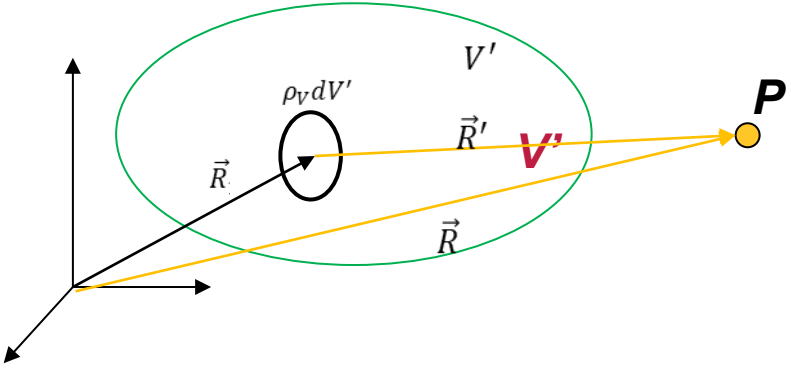
$$\nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t}\right) = 0 \quad \Rightarrow \quad \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla V$$

Electrodynamic (electric) potential

Electrostatic Potential Due to a Volume Charge Density

- The electrostatic potential at P is given by

$$V(\vec{R}) = \int_{V'} \frac{\rho_V(\vec{R}_i)}{4\pi\varepsilon_0 R'} dV'$$



Electrodynamic (Retarded) Potential Due to a Volume Charge Density

- The electrostatic potential at P is given by

$$V(\vec{R},t) = \int_{V'} \frac{\rho_V\left(\vec{R}_i, t - \frac{R'}{c}\right)}{4\pi\varepsilon_0 R'} dV'$$

It takes a finite time for a change in the source to cause an effect at the distant point.

The delay time depends on the speed of propagation denoted here as *c*.

Retarded Vector Magnetic Potential Due to a Volume Current Density

$$\vec{A}(\vec{R},t) = \frac{\mu}{4\pi}\int_{V'} \frac{\vec{J}\left(\vec{R}_i, t - \frac{R'}{c}\right)}{R'} dV'$$

Summary

- The derivatives of electromagnetic potentials can be used to obtain the electromagnetic field.
- In the time-varying case, retarded potentials that account for reaction time are used.

Ira A. Fulton Schools of Engineering

Arizona State University